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RESEARCH ARTICLE

Microbial Rhamnolipid-Stabilized mRNA Nanovaccines Enhance Adaptive Immunity via Dendritic Cell Targeting

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ABSTRACT

Poly(ethylene glycol) (PEG)-lipids endow lipid nanoparticles (LNPs) with colloidal stability but elicit anti-PEG immunity upon repeat dosing. Here it is replaced PEG-lipids with microbial rhamnolipids (RLs) to create PEG-free, self-adjuvanting LNPs. RL-LNPs show higher than 90% mRNA encapsulation and enhanced dendritic cell uptake via the mannose receptor and DEC205, redirecting expression to lymph nodes after intramuscular injection. Compared with size-matched PEG-LNPs, RL-LNPs enhance transfection and neutralizing capacity against pseudoviruses, while potently amplifying Th1-biased humoral and cytotoxic T-cell responses. Quantitative proteomics and western blotting reveal activation of the C-type lectin receptor (CLR)/NF-κB axis, corroborating intrinsic adjuvancy. Notably, RL-LNPs avoid PEG-associated mast-cell infiltration under repeat dosing while maintaining favorable systemic chemistry panels and histology. These data establish glycolipids as dual-function substitutes for PEG-lipids, coupling stabilization with receptor-programmed immunity for next-generation mRNA vaccines.

1 | Introduction

Lipid nanoparticles (LNPs) have emerged as indispensable delivery platforms for mRNA vaccines, exemplified by their pivotal role in COVID-19 vaccines and the recently authorized respiratory syncytial virus (RSV) vaccine mRNA-1345 [1–4]. Beyond mRNA vaccines, LNPs enable delivery of multiple nucleic acid modalities, including siRNA, and support in vivo T-cell engineering for CAR-T applications [5, 6]. LNPs typically comprise four key components, such as ionizable lipids for mRNA encapsulation, structural phospholipids and cholesterol to enhance membrane stability, and poly(ethylene glycol) (PEG)-conjugated lipids that confer colloidal stability and reduce macrophage clearance [7–10]. Notably, despite their clinical success, current LNP systems remain hindered by persistent challenges encompassing undesirable side effects and suboptimal immunostimulatory efficacy. These limitations have been partly attributed to the induction of anti-PEG antibodies during repeated administration [11–13]. Researchers observed that the injection of PEG-LNPs resulted in a notable elevation in the production of PEG-specific antibodies, with a 13.1-fold increase compared to the levels observed prior to vaccination [11]. The enhanced anti-PEG antibody level may induce hypersensitivity reactions by activating the complement system. Additionally, these antibodies can contribute to the accelerated blood clearance (ABC) effect, potentially reducing the therapeutic efficacy [14]. Therefore, this underscores the urgent need for alternative materials to replace PEG-lipids in LNP formulations.

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Emerging evidence from recent studies has explored synthetic lipopolymer hybrid systems as potential substitutes, achieving enhanced mRNA encapsulation efficiency and delivery performance while avoiding PEG-related immunogenicity [15, 16]. Previous studies have also incorporated polysorbate 80 into LNPs as a PEG-lipids alternative for siRNA delivery [17]. However, synthetic lipopolymers still suffer from limitations are hindered by notable limitations, including suboptimal biodegradability and undefined immunogenicity profiles [18]. Furthermore, polysorbate 80, despite its long history as a surfactant, exhibits intrinsic cytotoxic potential that requires careful toxicological evaluation [19]. These drawbacks highlight the need for biocompatible, biodegradable, and functionally versatile PEG-lipids alternatives.

Glycolipids are structurally characterized by a hydrophilic carbohydrate headgroup covalently linked to hydrophobic alkyl chains, conferring inherent amphiphilicity [20]. Such structural mimicry positions glycolipids as biologically derived alternatives for synthetic amphiphiles in nanocarrier design. As biosurfactants, they are environmentally friendly, generally less toxic, and can serve as natural analogues to synthetic amphiphiles in nanocarrier systems [21]. Furthermore, glycolipids constitute a class of biosurfactants that are perceived as more environmentally benign and less toxic [22]. Beyond structural competence in their physicochemical role, glycolipids exhibit intrinsic immunopharmacological activities. Acting as pathogen-associated molecular patterns (PAMPs), glycolipids can engage pattern recognition receptors on antigen-presenting cells, enabling dual functionality including nanoparticle stabilization and immune activation [22–24]. Compared to conventional adjuvants, glycolipid-based systems offer superior biocompatibility and reduced reactogenicity while maintaining potent immunostimulatory capacity [24]. Recent studies have highlighted synthetically modified glycolipids as promising alternatives to conventional PEG-lipids [25, 26]. Consequently, the strong industrial prospects of natural glycolipids, owing to their lower cost and enhanced feasibility for industrialization, have attracted significant research interest.

Rhamnolipids (RLs) are one of the most widely studied natural glycolipid biosurfactants, predominantly secreted by *Pseudomonas aeruginosa* [27]. These microbial-derived surfactants, like RLs, combine strong interfacial activity with biodegradability and documented immunomodulatory effects [28]. Notably, RLs enhance liposome stability and facilitate endosomal escape [29], which are two critical factors for effective mRNA vaccine delivery. These properties position RLs as promising PEG substitutes that can both stabilize LNPs and confer intrinsic adjuvant effects. Herein, we hypothesize that RLs can replace PEG-lipids in LNPs to achieve dual benefits of PEG-free nanoparticle stabilization and immune system activation.

In this study, we engineered RL-stabilized LNPs (RL-LNPs) and evaluated their performance in mRNA delivery and vaccination. Using luciferase mRNA (mLuc) and SARS-CoV-2 spike mRNA as model cargos, we assessed in vitro and in vivo transfection, innate and adaptive immune activation, safety profiles, and underlying mechanisms. This work establishes microbial glycolipids as a new class of multifunctional lipid components for next-generation mRNA vaccine design.

2 | Results and Discussion

2.1 | Development of RL-LNPs by Substituting PEG-Lipids

To generate PEG-free LNPs, RLs were incorporated in place of PEG-lipids. RLs are a class of mixed glycolipids with more than 60 distinct molecular structures, which can be categorized into 2 representative structural formulas (Figure 1a). The RLs used in this study are predominantly composed of Rha-$C_{10:0}$-$C_{10:0}$ (~30.82%), Rha-Rha-$C_{10:0}$-$C_{10:0}$ (~27.8%), and Rha-Rha-$C_{10:0}$ (~21.34%) (Figure S1), all characterized by saturated C10 fatty acid chains. This molecular architecture, shorter fatty chains, is thus proposed to facilitate their dissociation from nanoparticles during cellular uptake and actively promote the endocytic process [14, 30]. Furthermore, this disaccharide-glycosylated architecture aligns with prior reports suggesting that such motifs can enhance the transfection efficiency of nanoparticles [26]. A schematic highlights the replacement of PEG-lipids (ALC-0159) with RLs (Figure 1b). The initial RL-LNPs formulation was established based on the surfactant properties of RLs, drawing on a previously reported strategy employing polysorbate 80 [17]. Encapsulation of mLuc within RL-LNPs resulted in efficient transfection in HEK293 cells and in vivo (Figure S2), demonstrating that RLs, as amphiphilic glycolipids, can effectively stabilize nanoparticles and facilitate mRNA delivery. Formulation screening showed that increasing RL content reduced particle size and PDI while enhancing encapsulation efficiency (EE). RL-LNPs with 12 mol% RLs achieved maximal transfection (Figure S3a,b). Increasing ionizable lipids, enlarged particle size and reduced EE, but 25 % (mol/mol) gave optimal activity (Figure S3c,d).

The optimized formulation (28:14:46:12, ionizable lipid:DSPC:cholesterol:RL) exhibited a uniform particle size of ~158 nm, a PDI below 0.2, and an encapsulation efficiency exceeding 90%, while maintaining colloidal stability over a period of one week (Table S1 and Figure S4). TEM images and particle size distributions confirmed spherical and uniform morphology of both RL-LNPs and PEG-LNPs (Figure 1c). Compared with size-matched PEG-LNPs, DiD-labeled RL-LNPs showed markedly higher uptake in HEK293 cells (Figure 1d) and DC2.4 cells (Figure 1e). Flow cytometry confirmed more than a 2-fold increase in mean fluorescence intensity (MFI) compared to PEG-LNPs in both cell types (Figure 1f,g). Cellular uptake of RL-LNPs was clathrin-mediated in HEK293 cells and macropinocytic in DC2.4 cells (Figure S5). Lysosomal escape assays revealed a marked reduction in colocalization with lysosomes, validating the superior endosomal release capability of RL-LNPs. (Figure 1h–j). Nevertheless, the particle size of RL-LNPs is greater than that of launched PEG-LNPs (~90 nm), which may be attributed to the reduced nanoparticle stabilizing capacity of RLs in comparison to PEG-lipids. The larger size of the nanoparticles formulated with RLs may stem from fundamental structural differences between RLs and the standard PEG-lipids. Regarding the hydrophobic tail, the shorter chain length of RLs (~C10 alkyl) compared to those in commercial lipids could result in a less condensed core. Regarding the hydrophilic headgroup, studies indicate that the extent of glycosylation critically affects nanoparticle physicochemical properties and mRNA delivery efficiency [25]. The commercial PEG-lipid (ALC-0159), with

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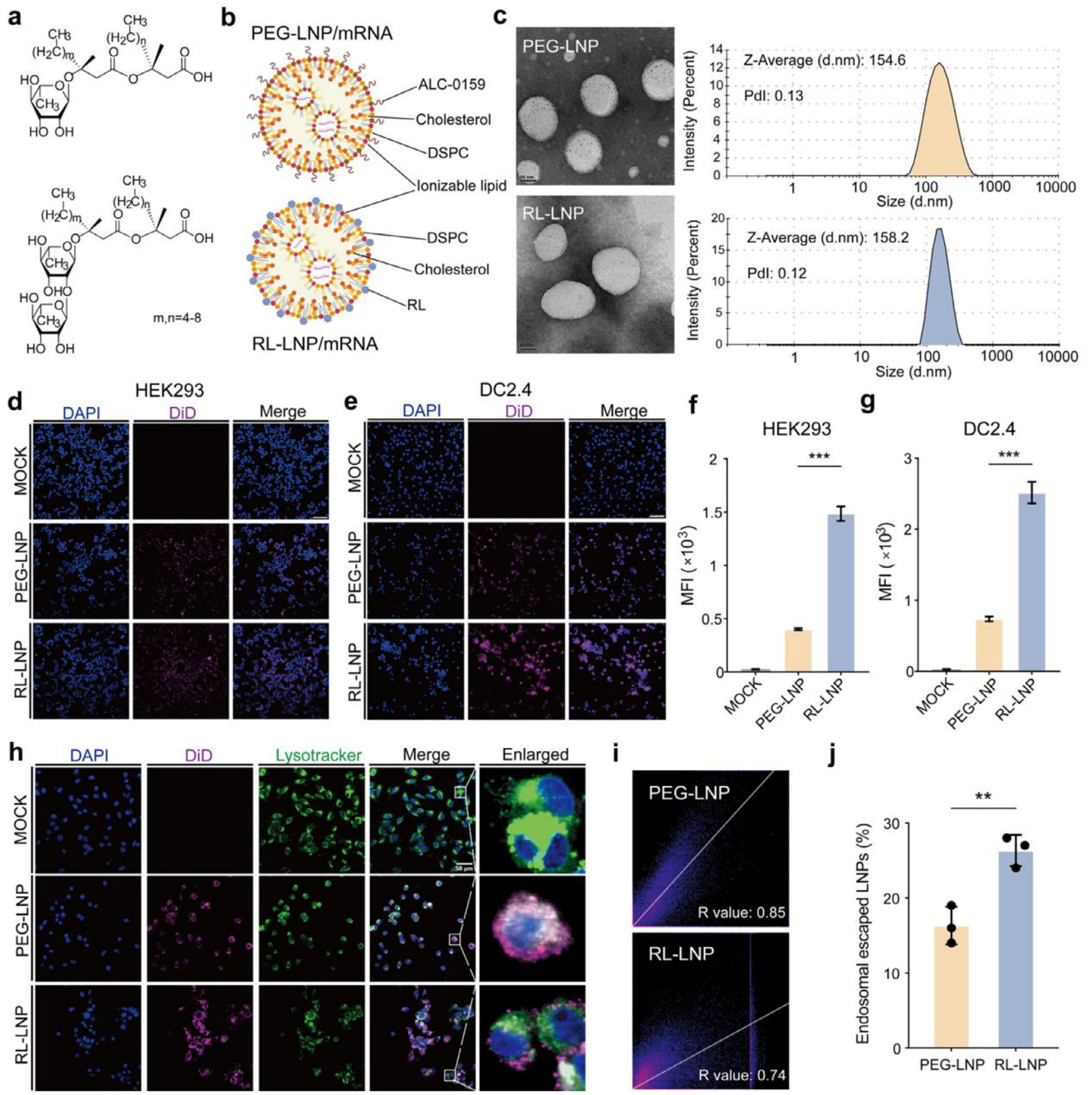


FIGURE 1 | Formulation, cellular uptake and lysosomal escape of PEG-LNPs and RL-LNPs. (a) Structures of common RLs. (b) Schematic diagrams of PEG-LNPs and RL-LNPs (created by biorender). PEG-LNPs are formulated with ionizable lipid, DSPC, cholesterol, and ALC-0159. RL-LNPs were formulated with ionizable lipid, DSPC, cholesterol, and RLs. (c) Representative TEM images and size distribution graph of PEG-LNPs and RL-LNPs. Scale bar, 100 nm. (d,e) Confocal images of HEK293 cells (d) and DC2.4 cells (e) treated with DiD-labelled (magenta) PEG-LNPs or RL-LNPs for 4 h. Nuclei were stained by DAPI (blue). Scale bars, 50 μm. (f,g) Flow cytometry analysis of HEK293 and DC2.4 treated with DiD-labelled PEG-LNPs or RL-LNPs for 4 h. MFI of HEK293 cells (f) and DC2.4 cells (g) were analyzed and presented (n=3). (h) Representative confocal images of DC2.4 cells treated with DiD-labelled (magenta) PEG-LNPs and RL-LNPs. Lysosomes were stained with lysotracker (green). Nuclei were stained by DAPI (blue). (i) Colocalization analysis of DiD and lysotracker. (j) Percentages of escaped nanoparticles were measured by Image J (n=3). Statistical significance was calculated via *t-test* in GraphPad Prism, $^{**}p < 0.01$, $^{***}p < 0.001$.

its 45-46 ethylene oxide units, provides strong hydrophilicity and steric repulsion. In contrast, the RLs used here, containing only one or two rhamnosyl groups, likely provide insufficient hydration and steric stabilization, potentially leading to larger particle sizes. Although larger than conventional PEG-LNPs, such sizes have been reported to enhance immune responses [31].

Previous studies have reported that the acyl chain of PEG-lipids must be shed before PEG-LNPs enter cells [32, 33]. However, RL-LNPs may bypass this process, thereby facilitating enhanced cellular uptake. Collectively, the results in Figure 1 attributes underpin the superior transfection performance of RL-LNPs, which will be evaluated in subsequent studies.

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2.2 | Improving Transfection and DC Maturation by RL-LNPs

The mLuc served as a model cargo to evaluate the delivery performance of RL-LNPs. Transfection with mLuc-loaded RL-LNPs induced a dose-dependent increase in luciferase expression in vitro (Figure S6a,b). Compared to PEG-LNPs, RL-LNPs exhibited an approximately 2-fold enhancement in luciferase expression in both HEK293 and DC2.4 cells (Figure 2a,b), consistent with their superior cellular uptake and lysosomal escape. Considering the critical role of DC activation in immune responses, we next assessed the capacity of RL-LNPs to promote DC maturation in vitro. SARS-CoV-2 spike mRNA-loaded RL-LNPs markedly enhanced bone marrow-derived dendritic cells (BMDCs) maturation in vitro, as reflected by a significant increase in $CD80^+CD86^+$ cells and elevated secretion of TNF-α, IL-1β, and IL-12p70 (Figure 2c,d), thereby highlighting their potential to potentiate downstream adaptive immune responses.

Subsequently, intramuscular administration of mLuc-loaded RL-LNPs elicited higher luciferase expression than PEG-LNPs (Figure 2e–g). While PEG-LNPs predominantly mediated luciferase expression in the liver, RL-LNPs induced a distinct distribution pattern, with most of the signal localized to the inguinal lymph nodes (iLNs) (Figure 2h,i). DiD labeling confirmed that both formulations remained largely in the muscle, with minimal liver or iLN retention (Figure S7). To assess their capacity for in vivo DC activation, mice were intramuscularly administered RL-LNPs or PEG-LNPs encapsulating SARS-CoV-2 spike mRNA as the model antigen. iLN cells were harvested and analyzed by flow cytometry, and cytokine levels were quantified by ELISA. Consistent with in vitro results, RL-LNPs increased activated DCs in iLNs by approximately 50%, accompanied by elevated TNF-α, IL-1β, and IL-12p70 levels compared to PEG-LNPs (Figure 2k,l), highlighting the superior DC-activating capacity of RL-LNPs in vivo [34]. Finally, to verify the absence of anti-RL antibody formation, we assessed in vivo luciferase expression following the first and the third administrations via both intramuscular (i.m.) and intravenous (i.v.) injection according to the method in the reported study [26]. The stable luciferase expression across repeated dosing regimens (Figure 2m,n; Figure S8) demonstrates that RL-LNPs evade anti-RL antibodies generation. Consequently, and unlike their PEGylated counterparts, RL-LNPs administered intravenously circumvent the antibody-mediated ABC phenomenon, highlighting their potential for repeated administration without loss of efficacy.

Consistent with previous reports that intramuscularly delivered mRNA vaccines are primarily internalized by local DCs [35, 36]. We propose that RL-LNPs follow a similar mechanism. They are likely internalized by muscle-resident DCs, and then these DCs migrate to iLNs. This also explains the absence of DiD-labelled nanoparticles in the iLNs and livers. DC maturation and activation mark the first step of the immune response and are essential for antibody production. Therefore, the in vivo immune response was further examined in detail.

2.3 | Humoral and Cellular Immunity Responses Induced by RL-LNPs

The adaptive immune responses elicited by PEG-LNPs and RL-LNPs were next evaluated. Mice were intramuscularly immunized three times at three-week intervals with nanoparticles containing SARS-CoV-2 spike mRNA (Figure 3a). At day 49 post-prime, serum antibodies, splenic B/T cells, and DCs in iLNs were analyzed. RL-LNPs induced significantly higher anti-SARS-CoV-2 spike antibodies than PEG-LNPs (Figure 3b left). To evaluate the functional potency of the humoral immune response elicited by RL-LNPs, a pseudovirus neutralization assay was performed using serum from immunized mice. The representative fluorescence images qualitatively demonstrate that serum from the RL-LNPs exhibited markedly superior neutralization capacity (Figure 3c). Quantitatively, 50% neutralization analysis confirmed that the RL-LNPs induced a substantially higher neutralizing antibody level, as reflected by a greater serum dilution factor (Figure 3d). Statistically, the RL-LNPs group achieved a significantly more potent response (dilution factor $\sim 3.5 \times 10^3$) compared to the PEG-LNPs group (dilution factor $\sim 1.2 \times 10^3$) (Figure 3e). Collectively, these data demonstrate that the RL-LNPs platform is superior in eliciting a robust and functionally potent neutralizing antibody response against SARS-CoV-2. Splenocytes were stimulated with spike peptide pools and assessed by flow cytometry. RBD-specific B cell activation in the spleen was markedly enhanced in the RL-LNPs group (Figure 3f). B cell responses are essential for antibody production, which accounts for the observed increase in serum antibodies. Total $CD4^+$ T cells were significantly increased (Figure 3g), with a pronounced rise in Th1 cells (Figure 3h) and comparable Th2 levels (Figure 3i), leading to an elevated Th1/Th2 ratio indicative of stronger cellular immunity and reduced allergic risk (Figure 3j,k). RL-LNPs also robustly activated splenic $CD8^+$ T cells, promoting differentiation into cytotoxic T lymphocytes (CTLs) (Figure 3l,m), and induced more pronounced DC activation in iLNs (Figure 3n). Confocal imaging confirmed increased $CD4^+$, $CD8^+$, and $B220^+$ cells in iLNs (Figure 3o,p), while IFN-γ levels were upregulated, reflecting enhanced antigen presentation (Figure S9).

Consistent with the enhanced lymphoid trafficking and luciferase expression observed in vivo (Figure 2), RL-LNPs elicited markedly stronger cellular immune activation compared to PEG-LNPs. As shown in Figure 3, RL-LNPs encapsulating SARS-CoV-2 spike mRNA induced significantly enhanced Th1 and $CD8^+$ T cell responses. The activation of Th1 cells provides protection against pathogens, which is beneficial in the context of viral infection and immunomodulation [37, 38]. The activation of $CD8^+$ T cells suggests the presence of both helper and cytotoxic functions, which are crucial for combating viral infections. Immune regulation involves a coordinated interplay of multiple signaling pathways and cytokine networks that collectively orchestrate adaptive immunity [39]. Cellular and humoral immunity act in concert to ensure comprehensive immune protection. Collectively, RL-LNPs uniquely potentiate adaptive immunity by synchronizing humoral (antibody/B cell), cellular (Th1/CTL), and innate (DCs) immune responses, establishing a balanced yet potent immunogenic profile. The potent immune-activating function of RL-LNPs prompted us to elucidate the mechanisms responsible for their distinctive immunomodulatory properties.

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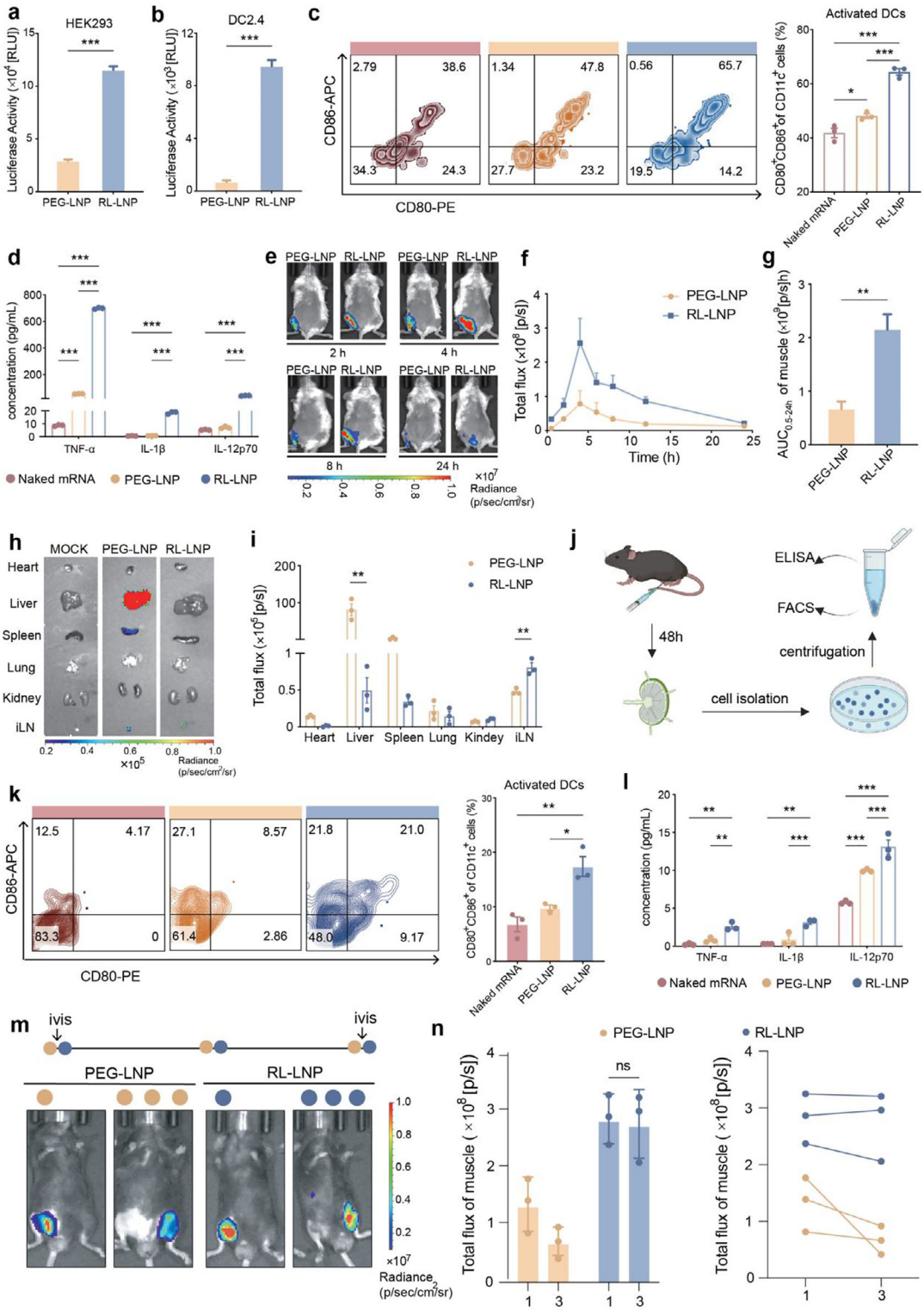


FIGURE 2 | Transfection and DCs activation. (a,b) Statistical data showed the luciferase activity of HEK293 cells (a) and DC2.4 cells (b) transfected with PEG-LNPs and RL-LNPs (n=3). (c) Representative flow cytometry diagram and statistical graph showing the maturation of BMDCs at 24 h after treatment of PEG-LNPs and RL-LNPs (n=3). (d) ELISA analysis of proinflammatory cytokines secreted by BMDCs, including TNF-α, IL-1β and IL-12p70 (n=3). (e) In vivo luminescence images at 2, 4, 8, and 24 h post-administration. (f) Statistical graph of luciferase expression in muscle (n=3). g $AUC_{0.5-24\,h}$ of muscle indicate total luciferase expression during 0.5–24 h (n =3). (h) Ex vivo luciferase distribution images of PEG-LNPs and RL-LNPs at 4 h after intramuscular injection. (i) Qualification of *ex vivo* luminescence signals in (h) (n=3). (j) A flow chart on analysis of DC activation in vivo (created by biorender). (k) Representative flow cytometry diagram and statistical graph showing the maturation of DCs in iLNs at 24 h after treatment of PEG-LNPs and RL-LNPs (n=3). (l) ELISA analysis of proinflammatory cytokines in LNs, including TNF-α, IL1β and IL-12p70 (n=3). (m) In vivo luminescence images at 6 h after the first administration and the third injection. (n) Quantification of luciferase expression in muscle (n=3). Statistical significance was calculated via *t-test* (a, b, g, i) or *one-way ANONA* (c, d, k, l, n) in GraphPad Prism, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

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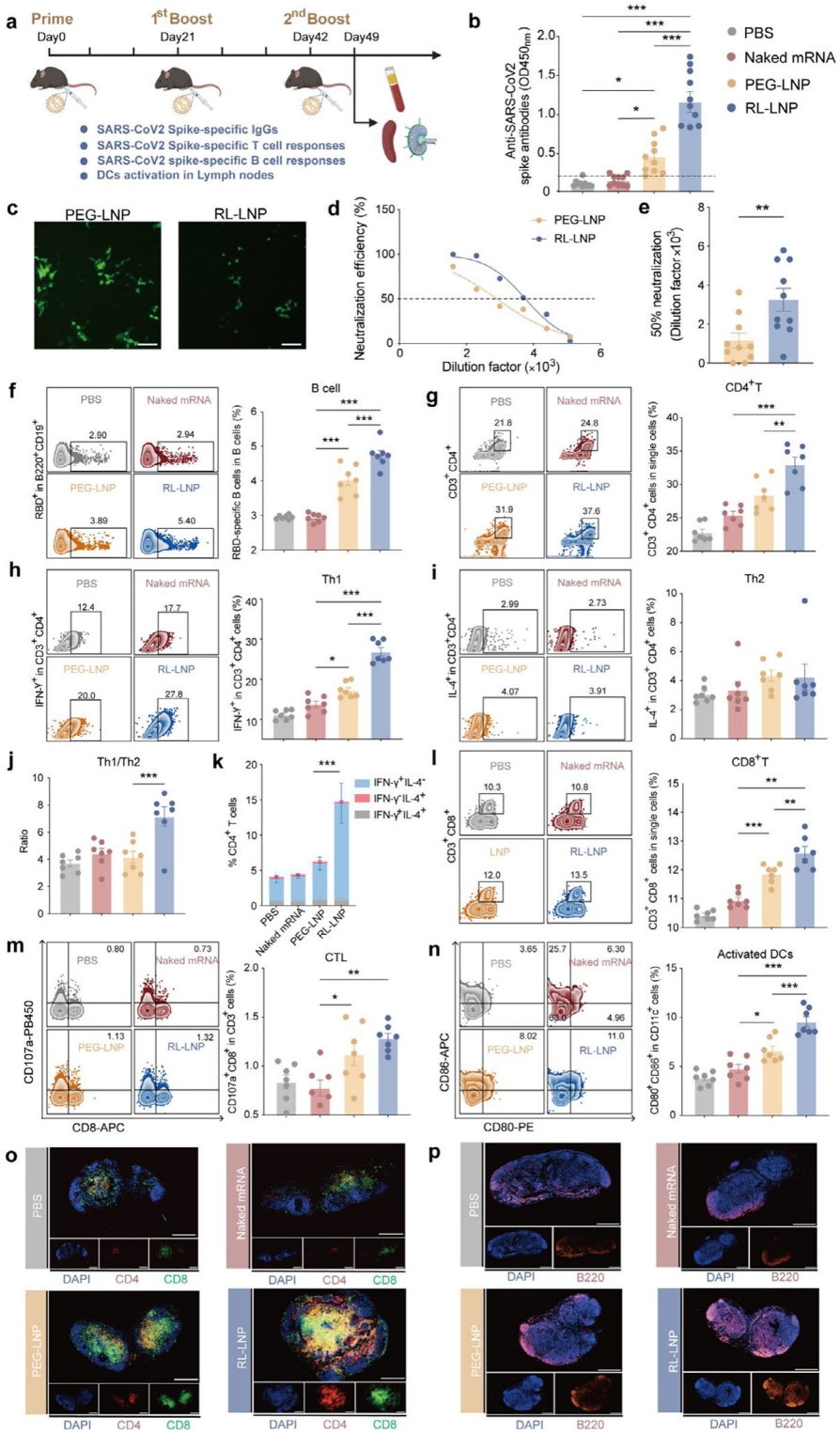


FIGURE 3 | Humoral and cellular immunity induced by RL-LNPs. (a) Schematic of the immunization and sample collection strategy (created by biorender). Serum, spleens, and LNs were harvested at 49 days post-initial vaccination. (b) ELISA analysis of anti-SARS-CoV-2-spike IgG antibodies (n=10). (c) Representative fluorescence images from a pseudovirus neutralization assay. Scale bars, 50 μm. (d,e) Quantitative neutralization assay of PEG-LNPs and RL-LNPs (n=10). (c) Flow cytometry analysis of SARS-CoV-2 spike-specific B cells (RBD$^+$B220$^+$CD19$^+$) in spleens (n=7). (d) Flow cytometry analysis of SARS-CoV-2 spike-specific CD4$^+$ T cells in spleens (n=7). (e) Percentage of Th1 cells (IFN-γ^+) among CD4$^+$ T cells (n=7). (f) Percentage of Th2 cells (IL-4$^+$) among CD4$^+$ T cells, (n=7). (g) Th1/Th2 cell ratio (n=7). (h) Frequency of polyfunctional CD4$^+$ T cells (n=7). (i) Flow cytometry analysis of SARS-CoV-2 spike-specific CD8$^+$ T cells in spleens (n=7). (j) Percentage of CTLs (CD107a$^+$) among CD8$^+$ T cells (n=7). (k) Flow cytometry analysis of activated DCs in LNs (n=7). (l,m) Representative fluorescence images showing CD4$^+$ T cells (red) and CD8$^+$ T cells (green) (l), and B cells (B220$^+$, red) (m) in LNs. Scale bar, 200 μm. Statistical significance was calculated via *one-way ANONA* in GraphPad Prism, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. Data in (b) below the dotted line were considered negative.

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2.4 | Immune Potentiation Mechanisms of RL-LNPs

RL is a class of glycolipids that have been regarded as immune adjuvants [40]. For example, globotriaosylceramide (Gb3), another glycolipid, has been reported to enhance humoral immune responses by regulating B cell function [41]. Based on this property, we hypothesized that RLs could function as a built-in adjuvant to enhance vaccine-induced immunity. To test this, SARS-CoV-2 spike mRNA-encapsulating PEG-LNPs (~150 nm) were co-administered intramuscularly with RL solution (Figure 4a). This combination significantly increased serum levels of anti-SARS-CoV-2 spike antibodies (Figure 4b). Moreover, it increased the frequencies of RBD-specific B cells, $CD4^+$ T cells, and $CD8^+$ T cells in the spleens (Figure 4c–e), as well as the proportion of activated DCs in the iLNs (Figure 4f). Collectively, RLs function as immune adjuvants that potentiate adaptive immune responses.

To investigate the enhanced immunity of RL-LNPs, we performed proteomic profiling in BMDCs, given their superior relevance to the in vivo immune milieu (Figure S10a). As illustrated in Figure S10b, compared to PEG-LNPs, RL-LNPs elicited a significantly different protein expression signature. Enrichment of membrane-associated proteins further reflected the initiation of signaling cascades (Figure S10c,d). CLR, a major immune receptor on DCs [42], was notably activated in BMDCs upon RL-LNPs treatment (Figure 4g). CLR pathway activation by RL-LNPs triggered downstream NF-κB signaling (Figure 4g). Consistently, RL-LNPs stimulation enhanced the phospho-p65 in BMDCs, confirming the activation of the NF-κB signaling pathway (Figure 4h). This resulted in elevated expression of IL-1, CXCL2, and other pro-inflammatory cytokines (Figure S10e,f). Gene Ontology (GO) analysis further confirmed the activation of cell surface signaling receptor pathways (Figure 4i; Figure S10g,h).

These results establish RLs as potent adjuvant-like alternatives to PEG-lipids. Their immunomodulatory activity originates from the glycosyl headgroup, which is supported by prior reports on glycopolymer adjuvants [43]. This property likely underlies the enhanced immunostimulatory activity of RL-LNPs. In addition, proteomic results are consistent with previous reports that CLRs can recognize glycolipids [43–45]. KEGG analysis and western blotting confirm that RL-LNPs activate immune responses via the CLR/NF-κB axis, thereby confirming the functional outcomes observed in Figure 3. Nonetheless, further mechanistic studies are required to fully elucidate their immunostimulatory properties.

2.5 | Specific Receptors Mediated Cellular Uptake of RL-LNPs

Building on initial observations indicating superior cellular uptake of RL-LNPs in DC2.4 cells compared to HEK293 cells (Figure 1d–g), we conducted a direct side-by-side comparison to validate this cell-selective internalization behavior. Results confirmed that RL-LNPs were internalized more efficiently by DC2.4 cells than by HEK293 cells (Figure 5a–c). In contrast, PEG-LNPs, which lack RLs, exhibited comparable uptake in both cell lines (Figure 5d). This selectivity is noteworthy, as HEK293 cells lacking specific receptors are generally highly permissive to nanoparticle transfection, suggesting that RL-LNPs entry is favored in DCs.

As professional antigen-presenting cells, DCs express diverse membrane receptors that facilitate antigen uptake and trigger immune responses [46]. It can thus be postulated that a specific receptor on the cell membrane of DC2.4 cells is responsible for mediating the uptake of RL-LNPs. To test this hypothesis, receptor competitive binding assays were conducted by co-incubating cells with DiD-labeled RL-LNPs and free RL solution. In DC2.4 cells, there was a notable decline in cellular uptake at an RLs concentration of 40 μg mL^{-1} (Figure 5e), whereas no such suppression occurred in HEK293 cells (Figure S11a). This cell-type-specific inhibition prompted us to further examine the phenomenon in RAW 264.7 macrophages, where RLs similarly suppressed nanoparticle uptake (Figure 5f). Receptors-mediated internalization of RL-LNPs by immune cells led to augmented uptake and trafficking, thereby explaining the enhanced and specific luciferase expression in LNs (Figure 2).

Previous data indicated that RL-LNPs activated an immune response through the C-type receptor pathway (Figure 4g). The mannose receptor (MR), a C-type lectin abundantly expressed on antigen-presenting cells, functions as both an endocytic and an immunostimulatory receptor capable of recognizing diverse glycostructures and facilitating their internalization [47, 48]. Consequently, we postulate that MR may serve as the specific receptor on DCs that mediates the cellular uptake of RL-LNPs. Receptor competitive binding assays revealed that mannose (40 μg mL^{-1}) markedly reduced RL-LNPs uptake in DC2.4 cells (Figure 5g). This inhibitory effect was recapitulated by RLs at equivalent concentrations in the same cell type (Figure 5h), but not in HEK293 cells (Figure S11b), indicating that MR mediates the recognition and endocytosis of RL-LNPs in immune-relevant cells. RLs unexpectedly enhanced nanoparticle uptake even in non-immune HEK293 cells (Figure S11a,b). This effect persisted when clathrin-mediated endocytosis or macropinocytosis was inhibited (Figure S11c). Since clathrin-mediated endocytosis and macropinocytosis are also the main uptake pathways of macrophages, a related mechanism may explain the modest uptake increase observed in RAW264.7 cells at low RL concentrations (Figure 5f). Specifically, RLs may activate downstream signaling in macrophages, promoting phagocytic vesicle formation and fusion to further facilitate RL-LNPs internalization [49]. Thus, RLs can modify endocytic pathways beyond specific receptors-mediated mechanisms. To further elucidate the specific endocytic receptors for RL-LNPs on DCs, we expanded our investigation to include additional candidate receptors. Besides the mannose receptor (MR, targeted by anti-CD206 antibodies), we also focused on DEC205 and Dectin-1, which are both conserved in human and murine DCs. Competitive binding assays using anti-CD206, anti-DEC205, and anti-Dectin-1 antibodies revealed that DEC205, but not Dectin-1, serves as another specific receptor mediating RL-LNP uptake in DCs (Figure 5i,j; Figure S12).

As summarized in Figure 5, the consistent inhibition of uptake across immune cell lines demonstrates that RL-LNPs exploit

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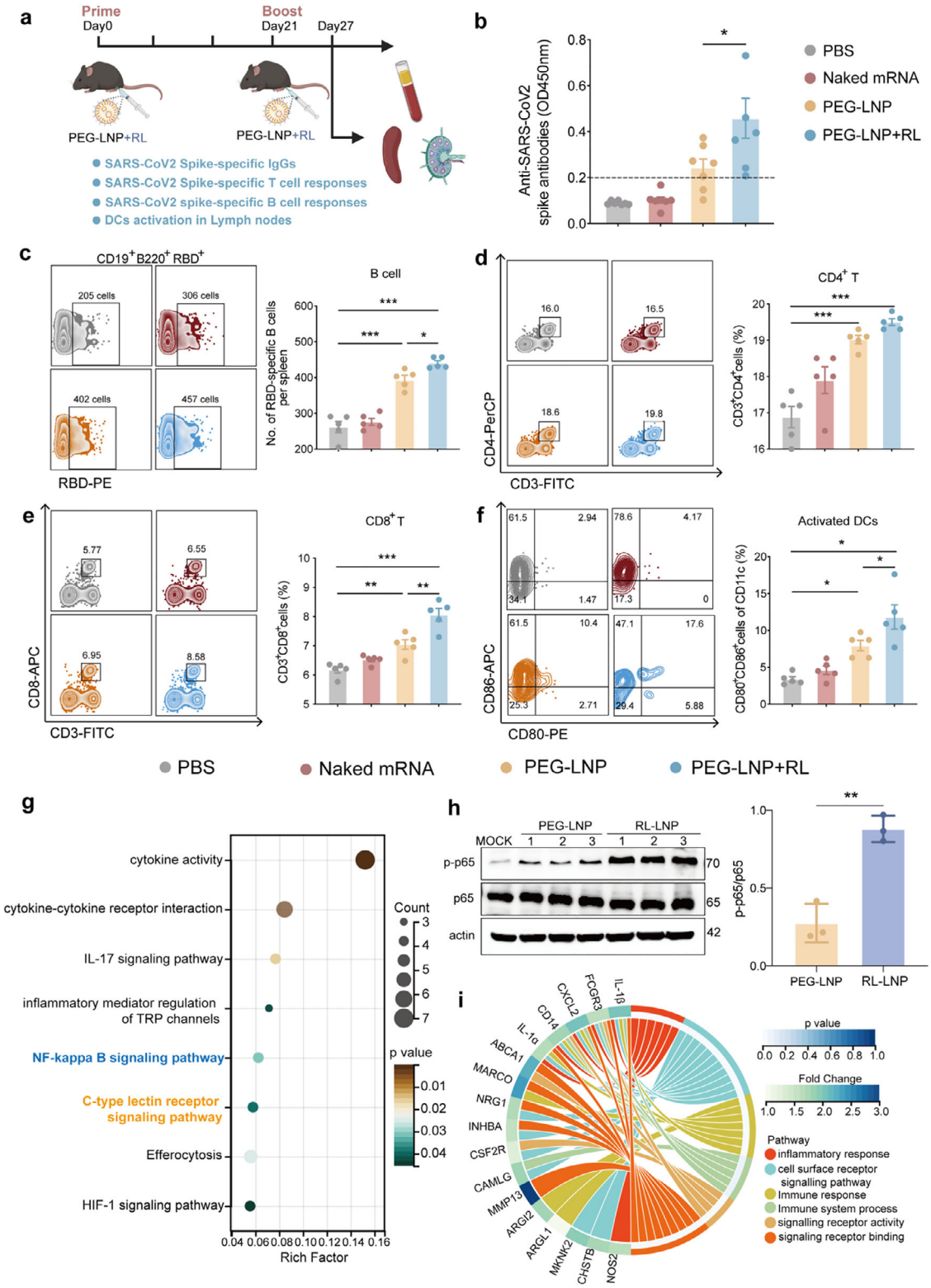


FIGURE 4 | Underlying mechanisms of RL-LNPs-mediated immune potentiation. (a) Schematic of the in vivo experimental timeline (created by biorender). Serum, spleens, and LNs were collected at 27 days after the first immunization. (b) ELISA analysis of anti-SARS-CoV-2 spike antibodies in serum (n=7). (c) Flow cytometry quantification of RBD-specific B cells in spleen (n=5). (d) Flow cytometry analysis of SARS-CoV-2-specific $CD4^+$ T cells in spleen (n=5). (e) Flow cytometry analysis of SARS-CoV-2-specific $CD8^+$ T cells in spleen (n=5). (f) Flow cytometry analysis of activated DCs in LNs (n=5). (g) KEGG pathway analysis of differentially expressed proteins in BMDCs transfected with PEG-LNPs and RL-LNPs. (h) Western blotting analysis of p-p65 and p65 in BMDCs treated with PEG-LNPs or RL-LNPs (n=3). (i) GO analysis of differentially expressed proteins in BMDCs transfected with PEG-LNPs and RL-LNPs. Statistical significance was calculated via t-test (h) or one-way *ANOVA* (b, c, d, e, f) in GraphPad Prism, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. Absorbance beyond 0.2 in (b) was considered positive.

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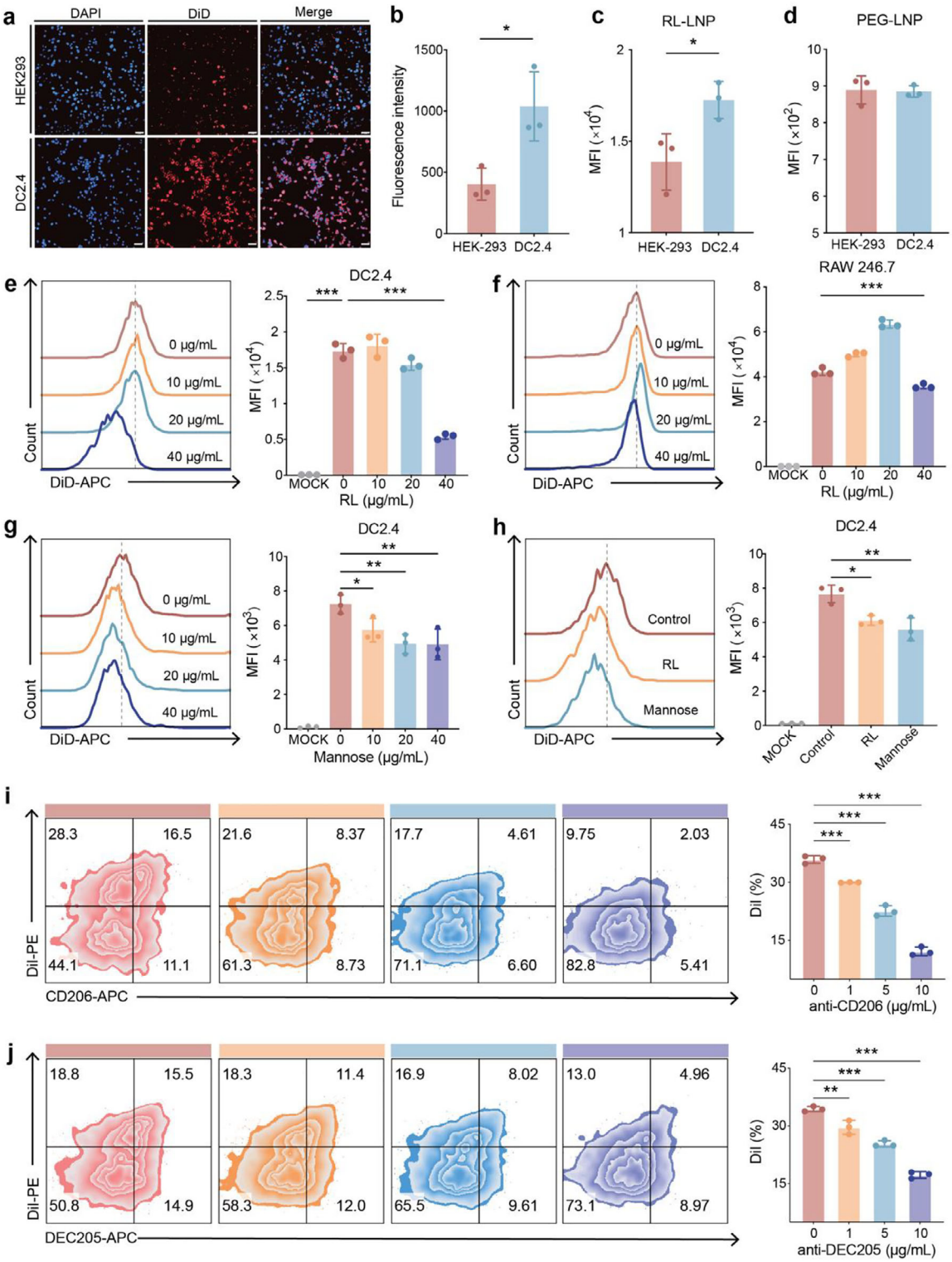


FIGURE 5 | Receptors-mediated cellular uptake of RL-LNPs. (a) Confocal images of DiD-labeled RL-LNPs uptake in HEK293 and DC2.4 cells. Scale bar, 50 μm. (b) Quantification of fluorescence intensity in (a) (n=3). (c) Flow cytometry analysis of DiD-labeled RL-LNPs uptake in HEK293 and DC2.4 cells (n=3). (d) Flow cytometry analysis of DiD-labeled PEG-LNPs uptake in HEK293 and DC2.4 cells (n=3). (e) Flow cytometry analysis of RL-LNPs uptake in DC2.4 cells pre-treated with varying concentrations of RLs (n=3). (f) Flow cytometry analysis of RL-LNPs uptake in RAW264.7 cells pre-treated with varying concentrations of RLs (n=3). (g) Flow cytometry analysis of RL-LNPs uptake in DC2.4 cells pre-treated with varying concentrations of mannose (n=3). (h) Flow cytometry analysis of RL-LNPs uptake in DC2.4 cells pre-blocked with RLs or mannose (n=3). (i) Flow cytometry analysis of DiI-labeled RL-LNPs uptake in DC2.4 cells pre-treated with varying concentrations of anti-CD206 antibodies (n=3). (j) Flow cytometry analysis of DiI-labeled RL-LNPs uptake in DC2.4 cells pre-treated with varying concentrations of anti-DEC205 antibodies (n=3). MOCK: untreated control; Control: no blocker added. Statistical significance was calculated via *t-test* (b, c) or one-way *ANOVA* (e, f, g, h, i, j) in GraphPad Prism, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

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common receptors, MR and DEC205, to achieve enhanced accumulation and internalization. Although macrophage uptake of RL-LNPs cannot be entirely ruled out, this does not preclude their targeted delivery to DCs. High CD206 expression is characteristic of M2 macrophages found in specific contexts like tumors, whereas resident macrophages in normal muscle express low mannose receptor levels [50, 51]. Critically, the luciferase signal detected in iLNs after i.m. injection (Figure 2h) is consistent with immune-cell (likely DC) uptake and transport of RL-LNPs. Our results highlight the potential of RLs as novel ligands in targeted delivery system design, enabling antibody-independent active targeting through their intrinsic biological recognition. Recognition of RL-LNPs is likely mediated by immune receptors beyond the identified mannose receptor and DEC205, potentially involving multiple CLRs that bind glycosylated motifs. A more comprehensive mapping of the specific receptors and pathways involved will be an important focus of future work.

2.6 | Safety Evaluation of RL-LNPs

Comprehensive safety assessments of RL-LNPs were subsequently conducted. To exclude immune-related interference, non-translated mRNA was employed in these evaluations. First, in vitro cytotoxicity assays were conducted. RL-LNPs exhibited no apparent cytotoxicity toward HEK293 or DC2.4 cells, as evaluated by cell viability assays (Figure S13). Next, a single-dose safety study was first performed at 48 h post-administration (Figure S14). The experimental design is illustrated in Figure S14a. Mice treated with PEG-LNPs exhibited a transient reduction in body weight at 48 h, whereas those receiving RL-LNPs maintained stable weight (Figure S14b). Moreover, the spleen and lymph node index were perturbed in the PEG-LNPs group but remained within normal physiological ranges in RL-LNPs-treated animals (Figure S14d,e). Serum biochemical analyses revealed that alanine aminotransferase (ALT), aspartate aminotransferase (AST), uric acid (UA), creatinine (CREA), creatine kinase-MB (CK-MB), and lactate dehydrogenase isoenzyme 1 (LDH-1) levels were within normal ranges for both PEG-LNPs and RL-LNPs treated mice (Figure S14f–k). Notably, proliferative abnormalities were observed in the livers of PEG-LNPs-treated mice (red arrow, Figure S14l), whereas RL-LNPs-treated livers appeared normal. Histological examination of muscle tissues at the injection site demonstrated preserved tissue architecture without evidence of damage or inflammation in both groups (Figure S14l).

Similar to the immune response study, a multi-dose safety evaluation was administered, resulting in stable body weight, liver index, and spleen index without significant alterations (Figure 6b–d). However, mice receiving PEG-LNPs exhibited an elevated lymph nodes index, whereas RL-LNPs-treated mice showed no significant alterations (Figure 6e). Biochemical analysis indicated that AST and ALT levels remained within the normal range in both groups, confirming the absence of liver dysfunction (Figure 6f,g). Serum UA levels showed a downward trend without statistical significance in the RL-LNPs group (Figure 6h), while a decrease in CREA was specifically observed in PEG-LNPs-treated mice but not in those receiving RL-LNPs (Figure 6i). CK-MB and LDH-1 levels exhibited normal profiles across all groups, indicating no cardiac toxicity (Figure 6j,k). Consistently, histological examinations revealed no pathological changes in the major organs of RL-LNPs-treated mice, whereas PEG-LNPs administration induced abnormal proliferative lesions in the liver, edema in iLNs, and nuclear introgression in muscle tissue (red arrows, Figure 6l). Importantly, mast cell infiltration was absent in RL-LNPs-treated tissues but was prominent in PEG-LNPs-treated mice (red arrows, Figure 6m). Furthermore, serum IgE levels were measured to assess potential allergic reactions, a known side effect associated with PEG-LNPs [52]. As shown in Figure S15, multiple injections of PEG-LNPs elicited a significant increase in IgE levels. In contrast, RL-LNPs did not induce such a response, indicating a favorable safety profile regarding allergenic potential.

Results in Figure 6e and i indicate that the elevated lymph nodes index observed in PEG-LNPs-treated mice was likely attributable to edema rather than immune cell proliferation. PEG-lipids in PEG-LNPs have been reported to elicit hypersensitivity reactions [35], consistent with the mast cell infiltration observed after repeated PEG-LNPs administration. In contrast, the lack of PEG-lipids in RL-LNPs precluded such effects, indicating that replacing PEG-lipids with RLs can mitigate the side effects associated with repeated PEG-LNPs dosing. The stable serum IgE levels in mice treated with RL-LNPs further corroborate the absence of an anti-RL antibodies response. Taken together, RL-LNPs demonstrated a favorable safety profile, supporting their potential as promising alternatives to PEG-lipids.

3 | Conclusions

In summary, this study introduces RLs as a novel, natural, and economical alternative to PEG-lipids, equipping lipid nanoparticles with dual functionality: colloidal stabilization and intrinsic immunostimulation. The resulting PEG-free RL-LNPs exhibited favorable physicochemical properties and significantly enhanced mRNA transfection efficiency. While the larger particle size of RL-LNPs (~150 nm) compared to clinically used PEG-LNPs (~90 nm) could, in principle, limit transfection efficiency [53]. In our current formulation, achieving both a small particle size and high encapsulation efficiency presents a recognized technical trade-off. To directly isolate the effect of particle size in our comparative studies, we prepared larger PEG-LNPs (~150 nm) using the same lipid composition as in clinically approved mRNA vaccines by adjusting the microfluidic flow rate and the alcohol-to-water ratio. This study established RL as a viable and innovative alternative to PEG-LNPs. Future work will focus on optimizing the chemical structure and formulation of RLs to engineer smaller, next-generation RL-LNPs, thereby potentially enhancing their delivery performance. Mechanistically, the RL-LNPs activate DCs via the CLR/NF-κB pathway, leveraging the inherent adjuvant properties of the RLs. Furthermore, RL-LNPs achieve targeted delivery through mannose receptor and DEC205-mediated uptake by DCs, promoting efficient accumulation in iLNs (Scheme 1). Intramuscularly administered RL-LNPs also demonstrated a compelling safety profile. Collectively, this work establishes RL-LNPs as a promising, self-adjuvanted mRNA delivery platform, underscoring the broad potential of glycolipid-based nanotechnology.

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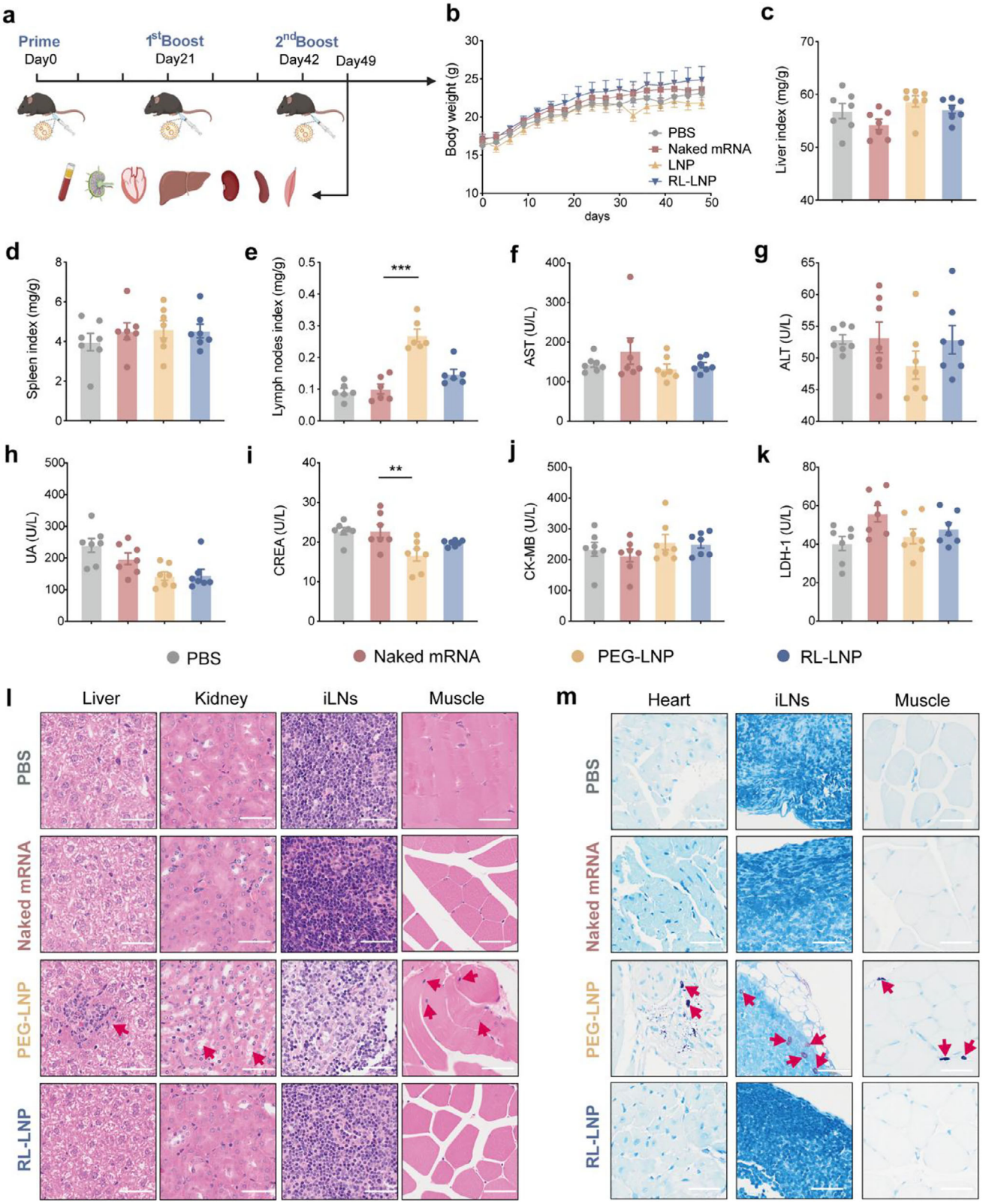
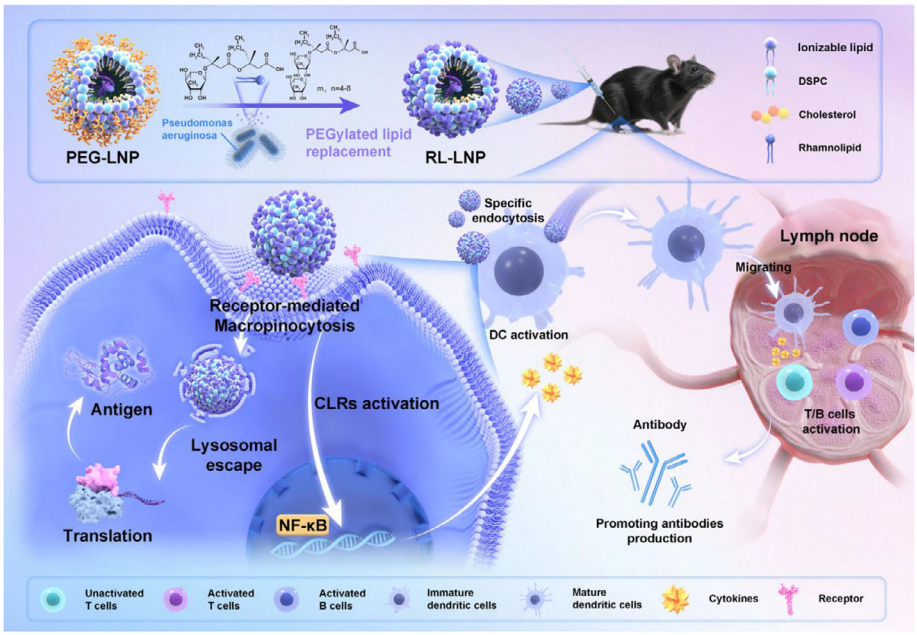


FIGURE 6 | Safety evaluation of RL-LNPs. (a) Schematic of the safety experimental timeline (created by biorender). Serum and major tissues were collected at 49 days after the first administration. (b) Body weight changes of mice in each group during 49 days. (c) Liver index (liver weight/body weight) at the endpoint (n=7). (d) Spleen index (spleen weight/body weight) (n=7). (e) Lymph nodes index (LNs weight/body weight) (n=7). (f–k) Serum biochemical indices: AST (f), ALT (g), UA (h), CREA (i), CK-MB (j), and LDH-1 (k) (n=6). (l) H&E staining of liver, kidney, iLNs, and muscle, scale bar: 50 μm. (m) Toluidine blue staining of heart, iLNs, and muscle. Red arrows indicate abnormal sites, scale bar: 50 μm. Statistical significance was calculated via one-way *ANOVA* in GraphPad Prism, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

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SCHEME 1 | Schematic illustration of RL-LNPs mRNA vaccine for enhanced immune activation. RLs, biosurfactants derived from *Pseudomonas aeruginosa*, serve as an alternative to PEG-lipids for stabilizing LNPs. Upon intramuscular administration, RL-LNPs are internalized by dendritic cells (DCs) via the specific receptors, facilitating activation through the C-type lectin receptor (CLR) signaling pathway. Mature DCs then migrate to lymph nodes, presenting antigens to naïve T and B cells, which triggers antigen-specific T cell expansion and antibodies production.

4 | Experimental Section

4.1 | Materials

RLs were purchased from Thermo Fisher Scientific (Waltham, USA). ALC-0315, DSPC, cholesterol, and ALC-0159 were purchased from AVT (Shanghai) Pharmaceuticals Tech Co., Ltd (Shanghai, China). mLuc and SARS-CoV-2 spike mRNA were purchased from APExbio (Houston, USA, Cat#R1013 and Cat#R1999). DiD and Lysotracker Green were purchased from Meilunbio (Dalian, China). Quant-iT RiboGreen RNA Reagent and LIVE/DEAD fixable dead cell stain kit were purchased from Thermo Fisher Scientific (Waltham, USA). FITC anti-mouse CD11c antibody, PE anti-mouse CD80 antibody, APC anti-mouse CD86 antibody and APC anti-mouse B220 antibody were purchased from BD Biosciences (Franklin Lakes, USA). GM-CSF and IL-4 were purchased from Yeasen biotech (Shanghai, China). TNF-α, IL-1β and IL-12p70 ELISA kits, anti-mouse CD19, anti-mouse CD16/32 and mouse Th1/Th2 staining kit were purchased from Lianke bio (Hangzhou, China). Mouse anti-SARS-CoV-2 spike IgG ELISA kit was purchased from Jonln bio (Shanghai, China). SARS-CoV-2 spike RBD peptide and Biotinylated SARS-CoV2 spike RBD probe were purchased from Sino biological (Beijing, China). PE streptavidin was purchased from Biolegend (San Diego, USA). Anti-mouse CD28 antibody and mannose were purchased from Medchemexpress (Shanghai, China).

4.2 | Nanoparticle Preparation

PEG-LNPs and RL-LNPs were formulated using microfluidics [54]. Lipids with specified molar ratios (as listed in Table S1) were dissolved in ethanol to form the organic phase, while mRNA was dissolved in citrate buffer (pH 4.0) to constitute the aqueous phase. The two phases were mixed using a microfluidic device at a volume ratio of 1:3 (organic to aqueous) with an N/P ratio of 10. The resulting lipid–mRNA mixture was subjected to ultracentrifugation at 1000 × g for 10 min at 4°C to remove residual ethanol.

4.3 | Nanoparticle Characterization

The particle size, PDI and zeta potential of PEG-LNPs and RL-LNPs were measured using a Zetasizer Nano (Malvern Instruments, Worcestershire, UK). The mRNA encapsulation efficiency was detected using a Quant-iT RiboGreen RNA assay. 0.5% TritonX-100 was used to lyse nanoparticles and to release mRNA. The morphology of PEG-LNPs and RL-LNPs was examined using transmission electron microscopy (TEM) with a JEM-1230 instrument (JEOL, Tokyo, Japan). Samples were negatively stained with phosphotungstic acid and placed onto copper grids. TEM imaging was performed at an accelerating voltage of 200 kV.

4.4 | Cell Culture

HEK293 and DC2.4 cell lines were purchased from American Type Culture Collection (ATCC). HEK293 cell line was maintained in Dulbecco's modified Eagle's medium (DMEM) with 10% FBS, 100 U mL^{-1} penicillin and 100 μg mL^{-1}. DC2.4 cell line was cultured in RPMI 1640 medium with 10% FBS, 100 U/mL penicillin and 100 μg mL^{-1} streptomycin. All cells were cultured at 37 °C, 5% CO_2. BMDCs were derived from the bone marrow cells of C57BL/6 mice. Briefly, bone marrow cells were flushed out using a 1 mL syringe from the femurs

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of mice and lysed using red blood cell lysis buffer. Then, cells were cultured in RPMI 1640 medium with 10% FBS, 100 U mL^{-1} penicillin and 100 μg mL^{-1} streptomycin, 0.1 mM β-mercaptoethanol, 20 ng mL^{-1} granulocyte-macrophage colony-stimulating factor (GM-CSF), and 10 ng mL^{-1} murine IL-4. The suspended and loosely adherent cells were collected on day 6.

4.5 | Cellular Uptake

HEK293 cells and DC2.4 cells were cultured in 12-well plates at a density of 5×10^5 cells per well overnight. Then, cells were treated with PEG-LNPs or RL-LNPs labeled with DiD at a mRNA density of 500 ng mL^{-1} for 4 h. Following that, cells were collected and washed twice using 1 × PBS. Cellular uptake was detected by flow cytometry and mean fluorescence intensity (MFI) were measured using FlowJo. HEK293 cells and DC2.4 cells were seeded into glass-bottom 24-well plates at a density of 2×10^5 cells per well overnight. After treating with DiD-labeled PEG-LNPs or RL-LNPs, cells were fixed using 4% paraformaldehyde (PFA) for 5 min. Cells were washed twice with 1 × PBS following staining with DAPI. Cellular uptake was also detected using confocal imaging. Fluorescence images were taken using a confocal laser scanning microscope.

4.6 | Lysosomal Escape

DC 2.4 cells were incubated with DiD-labeled PEG-LNPs or RL-LNPs, followed by staining with LysoTracker Green for 2 h. Cells were then fixed with 4% paraformaldehyde (PFA) and counter-stained with DAPI for 5 min. Confocal laser scanning microscopy was employed to visualize the fluorescence signals. Lysosomal escape efficiency was evaluated by quantifying the colocalization of DiD and LysoTracker Green signals using ImageJ software.

4.7 | Receptor Competitive Binding Assays

The receptor competitive binding assays were analyzed using a flow cytometer. HEK293 cells and DC2.4 cells were seeded onto 12-well plates overnight. Cells were treated with a combination of RL-LNPs (or PEG-LNPs) labeled with DiD/DiI and RLs (or mannose, or antibodies) for 4 h. Then, cells were collected and washed with 1 × PBS. For antibodies binding assays, cells were stained with anti-mouse CD206 antibodies, anti-mouse DEC205 antibodies or anti-Dectin1 antibodies for 30 min. Cellular uptakes were detected using a flow cytometer.

4.8 | In Vitro Transfection

HEK293 cells and DC2.4 cells were seeded into 24-well plates at a cell density of 2×10^5 cells for 24 h. And then, cells were treated with mLuc-loaded PEG-LNPs or RL-LNPs at an mRNA concentration of 500 ng mL^{-1} for 4 h in a serum-free environment. Subsequently, the medium was replaced with a serum-containing one, and cells were cultured for 24 h continuously. The luciferase expression was evaluated using a firefly luciferase reporter gene detection kit.

4.9 | In Vivo Transfection and Luciferase Distribution

All animal experiments were approved by the Institutional Animal Care and Use Committee (IACUC) at the School of Pharmacy, Fudan University, China, and all animals experiments were approved by Experimental Animal Ethics Committee of School of Pharmacy Fudan University (2024-06-YJ-QJP-82). All animals were housed in a specific pathogen-free (SPF) facility under controlled temperature and humidity conditions with a 12 h light/dark cycle. Mice were injected with mLuc-loaded PEG-LNPs or RL-LNPs at an mRNA density of 5 μg per mouse intramuscularly. Mice were administered intraperitoneal injections of D-Luciferin potassium salt (150 mg kg^{-1}) at 4 h post-injection. Subsequent imaging was performed using the in vivo imaging system (IVIS) Lumina system (PerkinElmer). Quantification of bioluminescence values was achieved by measuring photon flux (photons/second) within the region of interest using Living Image Software (PerkinElmer). Organs from injected mice were harvested at 12 h post-injection and imaged using IVIS. The value of total luminescence was measured and analyzed.

4.10 | BMDCs Maturation Analysis

BMDCs were cultured in a 12-well plate at a density of 1×10^6 cells per well for 24 h. Then cells were treated with SARS-CoV-2 spike mRNA-loaded PEG-LNPs or RL-LNPs at an mRNA concentration of 500 ng mL^{-1} in an environment of serum-free environment for 4 h. Later, the medium was replaced to a complete one and cells were cultured for 24 h continuously. Cells were collected and stained with FITC anti-mouse CD11c antibody, PE anti-mouse CD80 antibody and APC anti-mouse CD86 antibody for 30 min. Then cells were washed with 1×PBS and fixed with 2% PFA. Flow cytometry was utilized to detect the mature BMDCs ($CD80^+$ $CD86^+$). Meanwhile, the cell culture of transfected BMDCs was collected for ELISA. TNF-α, IL-1β and IL-12p70 were detected according to protocols of ELISA kits. The gating strategies were exhibited in Figure S16.

4.11 | In Vivo DCs Maturation Analysis

iLNs were dissected from immunized mice at 24 h post intramuscular injection of SARS-CoV-2 spike mRNA-loaded PEG-LNPs or RL-LNPs at a mRNA density of 5 μg per mouse. LNs were mechanically dissociated, and single cells were isolated. Then LN cells were blocked using anti-mouse CD16/32 antibody and stained with FITC anti-mouse CD11c antibody, PE anti-mouse CD80 and APC anti-mouse CD86 for 30 min. Flow cytometry was used to detect the mature DCs in LNs ($CD80^+$ $CD86^+$). Cells from LNs were cultured in a 96-well plate at a density of 1×10^4 cells per well for 8 h. Cell culture supernatant was collected and detected using ELISA of TNF-α, IL-1β and IL-12p70. The gating strategies for matured DCs in LNs were shown in Figure S16.

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4.12 | In Vivo Immunization

Mice were injected intramuscularly with SARS-CoV-2 spike mRNA-loaded PEG-LNPs or RL-LNPs at a density of 5 μg per mouse three times at a 3-week interval. Serum of immunized mice were collected at day 49 post-prime immunization. Anti-SARS-CoV-2 spike antibody in serum of immunized mice was detected and analyzed using an ELISA kit. Spleens and iLNs of immunized mice were harvested for the detection of T cells, B cells and DC cells.

4.13 | Pseudovirus Neutralization Assay

The pseudovirus-based neutralization assay was performed to evaluate the neutralizing activity of serum samples or antibodies. Briefly, serial dilutions of heat-inactivated test samples were incubated with a predetermined titer of SARS-CoV-2 pseudovirus for 1 h at 37 °C. Subsequently, the mixture was added to 96-well plates pre-seeded with ACE2-expressing HEK293T cells. Following incubation for 48–72 h at 37 °C with 5% CO_2, the culture medium was removed, and the luciferase activity in the cells was measured. Relative luminescence units were recorded with a microplate reader. Neutralization efficiency was calculated by determining the sample dilution that inhibited 50% of the pseudovirus infection compared to the virus control wells using non-linear regression analysis.

4.14 | Spleen T Cell Analysis

Spleens of immunized mice were harvested on day 49 after the primary injection, and single-cell suspensions were prepared using 70 μm cell strainers. Splenocytes were resuspended in red blood cell lysis buffer. A total of 2.5 μg mL^{-1} SARS-CoV-2 spike RBD peptide pools and 2 mg mL^{-1} anti-CD28 antibody were used to stimulate splenocytes (2×10^5 cells per well) in 24-well plates for 1 h. Subsequently, the cells were treated with 5 μg mL^{-1} brefeldin A, 2 mM monensin, and 5 μg mL^{-1} anti-CD107α antibody for an additional 5 h. After stimulation, splenocytes were washed with 1× PBS and blocked with anti-mouse CD16/32 antibody for 15 min. Cells were then stained using a mouse Th1/Th2 staining kit and fixed with 2% paraformaldehyde (PFA). In parallel, stimulated splenocytes were stained with anti-CD8 antibody and fixed. Flow cytometry was performed to analyze Th1/Th2 cell subsets and CTLs. Gating strategies for $CD4^+$ T cells, $CD8^+$ T cells, Th1, Th2, and CTLs were shown in Figure S17.

4.15 | Spleen B Cell Analysis

Spleens of immunized mice were harvested at day 49 post-prim immunization and filtered single cells with 70 μm cell strainers. Cells were lysed with red cell lysis buffer and washed with 1× PBS. Then cells were stained with mFluor 450 anti-mouse CD19 antibody, FITC anti-mouse B220 antibody and fluorescent SARS-CoV-2 spike RBD probe for 30 min. Following washing with 1× PBS, cells were fixed with 2% PFA. Then, stained cells were detected and analyzed using flow cytometry. Gating strategies for SARS-CoV-2-specific B cells in the spleen are exhibited in Figure S18.

4.16 | Proteomic Analysis

BMDCs were isolated from C57BL/6 mice as described in the method *Cell culture*. Purified BMDCs were then incubated with SARS-CoV-2 spike mRNA-loaded RL-LNPs or PEG-LNPs. After treatment, cells were harvested and lysed in buffer to a final concentration of 1% sodium deoxycholate (SDC) in 100 mM Tris–HCl (pH 8.5), followed by sonication. Lysates were clarified by centrifugation at 12 000 g for 5 min, and protein concentration in the supernatant was determined by BCA assay. Equal amounts of protein were adjusted to the same volume with 1% SDC in 100 mM Tris–HCl (pH 8.5). Reduction and alkylation were performed by adding TCEP and CAA and incubating at 60°C for 30 min. Samples were then diluted with an equal volume of ddH_2O to reduce SDC to < 0.5% before digestion with trypsin (enzyme: protein = 1:50, w/w) at 37°C with agitation overnight. Digestion was quenched with TFA, and peptides were clarified by centrifugation at 12 000 g. Peptides were desalted on self-packed SDB columns, vacuum-dried, and stored at −20 °C until analysis. Peptide samples were analyzed on an UltiMate 3000 RSLCnano system (Thermo Fisher) coupled to a timsTOF Pro mass spectrometer (Bruker) operating in diaPASEF mode. Approximately 200 ng of digest was loaded via an autosampler onto a C18 trap column (75 μm × 2 cm, 3 μm, 100 Å; Thermo) and separated on an analytical column (75 μm × 15 cm, 1.7 μm, 100 Å; IonOpticks). Mobile phases were 0.1% formic acid in water (A) and 0.1% formic acid in acetonitrile (B), run over a linear gradient. The capillary voltage was set to 1500 V. MS1 and MS2 scans were acquired from m/z 100–1700, and ion mobility ($1/K_0$) ranged from 0.6 to 1.6 Vs cm^{-2}. Accumulation and ramp times were 50 ms each. diaPASEF acquisition windows were defined in timsControl based on the m/z–ion mobility distribution. Collision energy was ramped linearly from 59 eV at $1/K_0$ = 1.6 Vs cm^{-2} to 20 eV at $1/K_0$ = 0.6 Vs cm^{-2}. Raw DIA data were processed in library-free mode using DIA-NN 1.8.1 against the UniProt mouse proteome (downloaded 2024-01-09; 55,086 entries). Search parameters were largely default, with precursor ion generation enabled; Trypsin/P specificity allowing up to one missed cleavage; carbamidomethylation (C) as fixed modification; oxidation (M) and N-terminal acetylation as variable modifications; MS1 and MS2 mass tolerances of 15 ppm; match-between-runs (MBR) and heuristic protein inference enabled; and a 1% FDR. Protein intensities were normalized using the MaxLFQ algorithm.

4.17 | Adjuvant Efficacy Evaluation of RL

Mice were immunized with PEG-LNPs or PEG-LNPs in combination with 20 μg mL^{-1} RL at a mRNA dose of 5 μg per mouse. Mice were administered biweekly for two doses. Serum, spleens and iLNs were harvested at day 28 post-prime administration. Anti-SARS-CoV-2 spike antibodies were detected using ELISA. Meanwhile, specific T cells, B cells in the spleens and DCs in iLNs were analyzed using flow cytometry.

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4.18 | Western Blotting

BMDCs were isolated from C57BL/6 mice as described in the method *Cell culture*. Purified BMDCs were then incubated with SARS-CoV-2 spike mRNA-loaded RL-LNPs or PEG-LNPs. Cells or tissues were lysed in RIPA buffer containing protease and phosphatase inhibitors at 48 h post-incubation, followed by centrifugation to collect the protein supernatant. Protein concentration was determined using a BCA assay, and samples were denatured in Laemmli buffer by heating at 95°C for 10 min. Equal amounts of protein were separated by SDS-PAGE on a 10% polyacrylamide gel using Tris-Glycine-SDS running buffer. The separated proteins were then transferred onto a PVDF membrane via wet transfer in Tris-Glycine-Methanol buffer at 400 mA for 60 min. After transfer, the membrane was blocked with 3% BSA in TBST for 1 h at room temperature, followed by incubation with primary antibody diluted in blocking buffer overnight at 4 °C. The membrane was washed with TBST and incubated with an HRP-conjugated secondary antibody for 1 h at room temperature. Following additional washes, protein bands were visualized using enhanced chemiluminescence substrate and imaged with a chemiluminescence detection system.

4.19 | Safety Evaluation

Following immunization, serum, hearts, livers, spleens, lungs, kidneys and muscle at the injection site of mice were harvested at 48 h and on day 49. Body weight and organ weight were recorded and measured. ALT, AST, UA, CREA, CK-MB, and LDH-1 in serum were detected at both 48 h and day 49. Organs were fixed with 4% PFA and subjected to hematoxylin and eosin staining (HE).

4.20 | Statistics

All in vitro data were expressed as mean ± standard deviation (SD), and in vivo data were presented as mean ± standard error of the mean (SEM). One-way analysis of variance (ANOVA) and a Dunnett's multiple comparisons test were utilized to analyze the data. Statistical significance was defined as $p < 0.05$. Each experiment was independently repeated at least three times, yielding consistent results and the representative dataset was shown.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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